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CALIBRATION PROCEDURE

Humidity Indicators & Climatic Enclosures — per EURAMET Calibration Guide No. 20

1. Scope & Application

This procedure applies to the calibration of relative humidity and temperature indicators, hygrometers, and climatic/temperature-humidity controlled enclosures, per EURAMET Calibration Guide No. 20, "Calibration of Temperature and/or Humidity Controlled Enclosures". It covers both the calibration of stand-alone humidity indicators by comparison, and the characterisation of chamber air conditions at defined locations within a climatic enclosure.

2. Reference Standards & Traceability

STANDARD	DESCRIPTION	TRACEABILITY
Reference Thermo-Hygrometer	Calibrated reference probe for temperature and relative humidity, spanning the test range	National metrology institute, ≤ 12 months
Humidity Generator / Chamber	Stable source of known temperature and relative humidity conditions	Characterised for stability and spatial uniformity

3. Environmental Conditions

PARAMETER	REQUIREMENT
Stabilisation	Chamber or generator allowed to reach and hold the set point before any reading is recorded
Probe Position	Reference and test probes positioned in close proximity, in the defined measurement location(s) within the chamber's useful volume
Loading State	Chamber loading condition (empty or loaded) recorded and held consistent with the client's intended use

4. Procedure

- Pre-Test Inspection.** Inspect the instrument or chamber for damage and confirm correct function across its operating range.
- Test Point Selection.** Select a minimum of 2–3 humidity set points (e.g. 30%RH, 50%RH, 80%RH) at one or more representative temperatures, per the client's stated requirement.
- Stabilisation.** At each set point, allow the chamber or generator to stabilise, confirmed by a steady reference reading over a defined dwell period.
Per cg-20 principle: calibration determines the deviation between the chamber's indicated value and the value measured by the reference at a defined position or for a defined useful volume.
- Comparison Reading.** Record the reference probe reading and the indication of the instrument or chamber display simultaneously at each set point, repeating a minimum of 3 times per point.
- Multiple Location Check (chambers only).** Where required, repeat measurements at multiple locations within the chamber's useful volume to characterise spatial uniformity.
- Calculate Error.** Determine the error as the indicated value minus the reference value at each test point, for both temperature and relative humidity.

5. Uncertainty & Calculation

The combined standard uncertainty includes: the calibration uncertainty of the reference thermo-hygrometer; the repeatability of readings at each set point; the spatial uniformity within the chamber's useful volume (where applicable); the temporal stability of the set point during the dwell period; and any radiative or self-heating effects on the reference probe. These are combined per the GUM method to give the expanded uncertainty ($k=2$) of the error at each test point.

6. Acceptance Criteria

The instrument or chamber passes calibration where the error at each test point, together with its uncertainty, falls within the manufacturer's specification or the client's stated tolerance requirement at the relevant measurement location(s).

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